AMA572 Lecture 1.

Credit Risk (I): Introduction and Default Probabilities

1 Introduction

Credit risk is the risk associated with default, which means that an obligor does not honor the payment obligations. As simple example, consider a defaultable bond which pays a face value (also called par value) \$1 at maturity T and has a continuous coupon payment ρ. Let the default time be τ. Assume the bond recovery value is $c(T,\tau)$; in other words upon default the bondholder can only get $c(T,\tau)$, which may be a random variable depending on the bond maturity T. Then the discounted total cash flow is

$$e^{-rT}\mathbf{1}_{\{\tau>T\}} + e^{-r\tau}c(T,\tau)\mathbf{1}_{\{\tau\le T\}} + \int_0^{\tau\wedge T}\rho e^{-rs}ds,$$

where r is the discount rate typically being the risk-free rate. The first term above is the discounted payoff if the default does not happen by time T, i.e. $\tau > T$. The second term is the payment upon default time τ. The last term is the coupon payment from time 0 until either the default time or the maturity, whichever comes first.

The default time τ may be triggered by various events, including bankruptcy (or filing for bankruptcy protection), failure to pay, etc. In reality, the coupon payment may not be continuous, e.g. semi-annual payments. The default payment $c(T,\tau)$ may result from legal actions and/or contractual obligations.

Credit risk is important for four reasons.

(1) Credit risk is relevant for defaultable bonds

Defaultable bonds include corporate bonds, sovereign bonds (issued in currencies other than domestic currencies). Because of the default risk, the yield for defaultable bonds will be higher than otherwise identical default-free bonds.

The yield (or yield to maturity) $v(T)$ is defined to be the equivalent $T-$year rate matching the bond price $B(0,T)$, i.e. $v(T)$ solves the equation

$$B(0,T) = e^{-v(T)T} + \int_0^T \rho e^{-v(T)s}ds.$$

The first term above is related to the payment at maturity, and the second term is related to the coupon payment. Since the r.h.s. of the above equation

is a decreasing function of $v(T)$, it has a unique solution. Also note that

$$\int_0^T \rho e^{-v(T)s} ds = \frac{\rho}{v(T)} \left(1 - e^{-v(T)T}\right),$$

so the equation can be rewritten as

$$B(0,T) = e^{-v(T)T} + \frac{\rho}{v(T)} \left(1 - e^{-v(T)T}\right).$$

Due to the default risk, the yield for defaultable bonds $v(T)$ will be bigger than the default-free rate r. The difference

$$v(T) - r \geq 0$$

is called yield spread. Yield spreads may have variety shapes, upward, downward, humped. Many factors may influence yield spreads, e.g. credit ratings, interest rate, etc. The shapes of credit spreads will be discussed in Lectures 3 and 4.

(2) Credit risk is relevant for the capital structure of a firm

Two ways for a firm to get capital are issuing bonds (or taking debts) or issuing stocks (or equity). The main differences between a bond and a stock are: (a) The equity holders control the firm. (b) Upon default the debt holders has the right to collect the money first, which means that the equity value is zero upon default.

One important corporate decision for a firm to make is to choose the debt equity ratio. In other words, if the firm needs money, the firm has to decide how much capital should be financed by issuing debt and how much capital should be financed by issuing equity. Credit risk plays an important role in deciding the capital structure. The higher probability of default, the more costly it is to issue debt, because of the potential bankruptcy costs. On the other hands, firms typically get tax benefits from the coupon payments of debts, as the coupon payments are tax deductible. Therefore, the choice of capital structure is a balance between the trade-off of the tax benefit and bankruptcy costs. The celebrated Modigliani-Miller theorem provides a framework to address this issue. We will discuss the capital structure in Lecture 2.

(3) Credit risk is relevant for the equity prices

As we just mentioned that upon default the equity value is zero. Thus, the equity values is affected by the credit risk. This in turn will affect options on the equity values. For example, on Jan 18th, 2006, the GM stock is traded at about \$20. The Jan 2007 put option with the largest open interest is

at the strike price 10, which is traded around $2.10. This implies a quite significant default probability for GM. Indeed GM went into bankruptcy during the 2008 financial crisis. This will be discussed in Lecture 5.

(4) Credit risk is relevant for pricing credit derivatives

Credit derivatives are useful for transferring credit risk from one party to another. Some popular credit derivatives are

- Credit default swaps (CDS), which pay certain amount once the default event occurs.
- Collateralized debt obligations (CDO), whose payoffs depend on the default events within a portfolio of bonds or debts.
- Credit spread options.

There are two basic modeling approaches to study credit risk. The first one is the structural approach, which attempts to first build models for the firm asset, and then study the implication of credit events to bond prices, equity values, options on bonds, and credit derivatives. The second one, reduced-form models, takes the bond prices (or more precisely the yield curves of bonds) as given in the market, and then try to study bond options and credit derivatives. In other words, the main difference between the two is that in the structural approach the bond prices are derived, while in the reduced-form approach the bond prices are taken from the market data. The structural models will be discussed in Lectures 2, 3, 4, 5, and the reduced-form models will be studies in Lectures 6 and 7.

2 Basic Terminologies

2.1 Credit Ratings

There are two major credit rating agencies in U.S., Moody's and S&P, which provide ratings indicating the creditworthiness of corporate bonds and of sovereign countries.

In Moody's system, the highest rating is Aaa, followed by Aa, A, Baa, Ba, B, and Caa. Bonds with ratings Baa or higher are called investment grade bonds, otherwise, junk bonds. The corresponding S&P ratings are AAA, AA, A, BBB, BB, B, and CCC. The finer ratings also exist. For example, for Moody's there are three subcategories within Aa, namely Aa1, Aa2, and Aa3, with Aa1 being higher than the other two; for S&P the subcategories with AA are AA+, AA, AA-.

Generally, the higher credit rating the lower probability of default. Below is the historical default probability from Moody's.

Average Cumulative Default Probability (in percent), 1970-2003. (From Moody's)

Years	1	2	3	4	5	7	10	15	20
Aaa	0.00	0.00	0.00	0.04	0.12	0.29	0.62	1.21	1.55
Aa	0.02	0.03	0.06	0.15	0.24	0.43	0.68	1.51	2.70
A	0.02	0.09	0.23	0.38	0.54	0.91	1.59	2.94	5.24
Baa	0.20	0.57	1.03	1.62	2.16	3.24	5.10	9.12	12.59
Ba	1.26	3.48	6.00	8.59	11.17	15.44	21.01	30.88	38.56
B	6.21	13.76	20.65	26.66	31.99	40.79	50.03	59.21	60.73
Caa	23.65	37.20	48.02	55.56	60.83	69.36	77.91	80.23	80.23

We can compute probabilities of default for a given year by using the above cumulative probabilities. For example, for Caa bonds, the average probability of default during year 2 is $0.3720 - 0.2365 = 0.1355$.

One empirical observation is that for high grade bonds the default probability is a increasing function of time, while for low grade bonds the default probability decreases over time. For example, for A-rated bonds, the average default probabilities during years 1, 2, 3, 4, 5 are 0.02%, 0.07%, 0.14%, 0.15%, and 0.16%, respectively; while for Caa rated bonds, they are 23.65%, 13.55%, 10.82%, 7.54%, and 5.27%. The reason is mean-reverting, i.e. a firm with great creditworthiness now will likely be less financially healthy later, and a firm facing financial distress will likely to improve its credit rating if the firm can survive during the first one or two years.

There are some connections between default probabilities and business cycles, especially for low-grade bonds.

2.2 Default Intensities

The previous table reported unconditional probabilities of default. For our purpose it is more convenient to study conditional probability of defaults, which is also related to hazard rate or default intensity. This is also useful to give us an intuition about the generator of a transition matrix later when we discuss how to estimate transition matrices.

The default intensity (or hazard rate) $\lambda(t)$ is defined to be

$$\lambda(t) = \frac{f_\tau(t)}{P(\tau > t)},$$

where τ is the default time, and $f_\tau(t)$ is the density function of τ.

To interpret $\lambda(t)$, consider the probability of default right after time t, given the default has not happened by time t, or more precisely $P(\tau \in (t, t+dt)|\tau > t)$. Now

$$P(\tau \in (t, t+dt)|\tau > t) = \frac{P(\tau \in (t, t+dt))}{P(\tau > t)} \approx \frac{f_\tau(t)dt}{P(\tau > t)} = \lambda(t)dt.$$

Thus, the default intensity gives the instantaneous conditional default probability.

To get an explicit representation of the hazard rate, note that

$$\lambda(t) = \frac{\frac{d}{dt}P(\tau \le t)}{1 - P(\tau \le t)}.$$

Integrating both sides gives

$$\int_0^s \lambda(t)dt = \int_0^s \frac{\frac{d}{dt}P(\tau \le t)}{1 - P(\tau \le t)}dt + C = \int_0^s \frac{dP(\tau \le t)}{1 - P(\tau \le t)} + C = -\ln(1 - P(\tau \le s)) + C,$$

or

$$P(\tau > s) = e^C \cdot \exp\left\{-\int_0^s \lambda(t)dt\right\}.$$

Putting $s = 0$ yields $1 = e^C$, and therefore

$$P(\tau > s) = \exp\left\{-\int_0^s \lambda(t)dt\right\}. \tag{1}$$

In terms of estimate the intensity from data, one can assume $\lambda(t)$ is piecewise constant, and replace $\frac{d}{dt}P(\tau \le t)$ by $\{P(\tau \le t + \Delta t) - P(\tau \le t)\}/\Delta t$. For example, for the B-rated bond, the estimator for $\lambda(2)$ will be

$$\frac{20.65 - 13.76}{100 - 13.76} = 0.07989.$$

3 Credit Transition Matrix

	Ratings at Year End							
Initial Rating	AAA	AA	A	BBB	BB	B	CCC	D
AAA	93.06	6.29	0.45	0.14	0.06	0.00	0.00	0.00
AA	0.59	90.99	7.59	0.61	0.06	0.11	0.02	0.01
A	0.05	2.11	91.43	5.63	0.47	0.19	0.04	0.05
BBB	0.03	0.23	4.44	88.98	4.70	0.95	0.28	0.39
BB	0.04	0.09	0.44	6.07	82.73	7.89	1.22	1.53
B	0.00	0.08	0.29	0.41	5.32	82.06	4.90	6.95
CCC	0.00	0.00	0.31	0.63	1.57	9.97	55.82	31.58
D	0	0	0	0	0	0	0	100

The above table is the historical worldwide corporate average one-year transition probabilities from 1980 to 2002, reported by S&P.

Clearly we can use the Markov chain method to analysis the above table. For example, (a) D is an absorbing state in the above Markov chain; (b) each row sum must equal to one; (c) if we denote the above matrix by $P = P_{(1)}$ for the one-year transition matrix, then the two-year transition matrix is $P_{(2)} = P^2$, by Chapman-Kolmogorov theorem.

4 Estimation of Credit Transition Matrix

There are at least three ways to do the estimation: empirical estimation, generator-based estimation, and intensity-based product estimation. Here are we discuss the first two methods. The last one is slightly more complicated and is valid for broader classes (e.g. non-homogeneous Markov chains with possible truncations in the observation).

As a illustration, consider the following example.

Example. (a) There are four credit classes, A, B, C, D. At the beginning, there are 10 firms in A, B, C, each.

(b) In one month, one B rated firm migrates to C, and one B rated firm migrates to A, and both firms stay there for the rest of the year.

(c) In two months, two A rated firms migrates to B, and stay there for the rest of the year.

(d) In six months, one C rated firm migrates to B and stays there for the rest of the year. Furthermore, one C rate firm defaults.

(e) Eight months later, a C rated firm defaults.

We can summarize the above transition in a table.

Months	1	2	3	4	5	6	7	8	9	10	11	12
A		B,B										
B	A,C											
C						B,D		D				
D												

4.1 Empirical Estimator

A simple way to estimate the one-year transition matrix is available by counting empirical transitions between states. For example, the empirical estimator for the transition probability from A to B is 2/10=0.2, as there are two A-rated firms migrate to B; similarly the empirical estimator for the transition probability from A to C and from A to D are all zero, as there is no such transition in the observation.

In summary, we have the following empirical estimator for the one-year transition matrix

$$\hat{P}_{empirical} = \begin{pmatrix} 0.8 & 0.2 & 0 & 0 \\ 0.1 & 0.8 & 0.1 & 0 \\ 0 & 0.1 & 0.7 & 0.2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

However, this estimator is not very good for several reasons.

(1) The estimator does not fully use the available information. For example, for this estimator, it does not matter whether the two transitions from A to B happen in two months or four months, or one happens in one month and another in five month. In other words, the exact time points of the transition are ignored in the estimation.

(2) The estimated matrix has too many zeros. For example, the estimated transition probability from A to C is zero. Since there is a positive probability from A to B (0.8) and a positive probability from B to C (0.1), we would like to have a positive probability from A to C. But since there is no observed transition from A to C, the estimator simply ignore this possibility.

Better estimation can be obtained by using other methods.

4.2 Generator-Based Estimator

We have shown that there is a one-to-one correspondence between default probability and default intensity. Similarly, according to the theory of Markov chain, there is a generator matrix Λ associated with the matrix P of transition probabilities between states. More precisely,

$$\Lambda = \begin{pmatrix} -\lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,K} \\ \lambda_{2,1} & -\lambda_{2,2} & & \lambda_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{K,1} & \lambda_{K,2} & & -\lambda_{K,K} \end{pmatrix}$$

and the sum of each row is equal to zero, i.e. $\lambda_{1,1} = \lambda_{1,2} + \cdots + \lambda_{1,K}$, etc. The intuition behind the generator Λ is that it gives the instantaneous transition matrix, just as what the hazard rate does for a random variable. Thus, it can be viewed as the hazard rate for the whole transition matrix. The element $\lambda_{i,j}$ measures the intensity from state i to state j.

Furthermore, similar to the relationship in (1) between a default probability and its intensity, the n-step transition matrix is simply the exponential of Λn

$$P^{(n)} = P^n = \exp\{\Lambda n\},$$

where for any matrix A the exponential function is defined to be

$$\exp\{A\} = \sum_{i=0}^{\infty} \frac{A^i}{i!}.$$

Most of software package can compute the above exponential function easily. For example in Matlab the code is `expm`. Note that the exponential function for a matrix is not equal to exponentiate each cells inside the matrix.

Since $\lambda_{i,j}$ measures the intensity from state i to state j, we can estimate it by

$$\hat{\lambda}_{i,j} = \frac{M_{i,j}}{N_i},$$

where $M_{i,j}$ is the number of transitions from i to j, and N_i is the cumulative time units in state i for all firms.

For example, in our example

$$N_A = 8 * \frac{12}{12} + 2 * \frac{2}{12} + 1 * \frac{11}{12},$$

as there are eight firms stay in state A the whole year, two firms stay in A for two months (then they move to B) and one firm stays in A for 11 months (as it moves from B to A after one month).

$$M_{A,B} = 2,$$

as there are two transitions from A to B.

In our example, the estimator becomes

$$\hat{\lambda}_{A,B} = \frac{2}{8 * \frac{12}{12} + 2 * \frac{2}{12} + 1 * \frac{11}{12}} = 0.21622,$$

$$\hat{\lambda}_{B,A} = \frac{1}{8 * \frac{12}{12} + 2 * \frac{1}{12} + 2 * \frac{10}{12} + 1 * \frac{6}{12}} = 0.09677\,,$$

$$\hat{\lambda}_{B,C} = \frac{1}{8 * \frac{12}{12} + 2 * \frac{1}{12} + 2 * \frac{10}{12} + 1 * \frac{6}{12}} = 0.09677,$$

$$\hat{\lambda}_{C,B} = \frac{1}{7 * \frac{12}{12} + 1 * \frac{11}{12} + 2 * \frac{6}{12} + 1 * \frac{8}{12}} = 0.10435,$$
$$\hat{\lambda}_{C,D} = \frac{2}{7 * \frac{12}{12} + 1 * \frac{11}{12} + 2 * \frac{6}{12} + 1 * \frac{8}{12}} = 0.20870,$$
$$\hat{\lambda}_{A,C} = \hat{\lambda}_{A,D} = \hat{\lambda}_{B,D} = \hat{\lambda}_{C,A} = 0.$$

Thus, the estimator of the generator matrix is given by

$$\hat{\Lambda} = \begin{pmatrix} -0.216\,22 & 0.216\,22 & 0 & 0 \\ 0.09\,677 & -0.19354 & 0.09\,677 & 0 \\ 0 & 0.104\,35 & -0.31305 & 0.208\,70 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Therefore, the one-year transition matrix is estimated by

$$\hat{P}_{generator} = \exp(\hat{\Lambda}) = \begin{pmatrix} 0.81407 & 0.17707 & 0.00825 & 0.00061 \\ 0.07925 & 0.83663 & 0.07556 & 0.00857 \\ 0.00398 & 0.08148 & 0.73507 & 0.17948 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The generator-based estimator uses the available information and leads to nonzero transition probabilities. Therefore, in this sense, it is better than the empirical estimator.

Exercise 1. Compute the empirical and generator-based estimators of one-year transition matrix for the following data.(a) There are four credit classes, A, B, C, D. At the beginning, there are 20 firms in A, B, C, each.

(b) In one month, one B rated firm migrates to C, and two B rated firms migrate to A, and both firms stay there for the rest of the year.

(c) In three months, one A rated firm migrates to B, and stays there for the rest of the year.

(d) In seven months, two C rated firms migrate to B and stay there for the rest of the year. Furthermore, one C rate firm defaults.

(e) Ten months later, two C rated firms default.

5 Differences between Historical Default Probabilities and Implied Risk-Adjusted Default Probabilities

Just like in the option pricing theory, one can get implied volatility from call/put options, i.e. implied risk-neutral volatility, or one can estimate historical volatility from historical stock returns. If (i) the market follows exactly geometric Brownian motion (ii) if the market is perfect (meaning no transaction costs, no borrowing and short-selling constraints, etc.) (iii) if one can trade the underlying asset continuously, then in theory the two approaches should give the same answer. However, in practice, none of the above conditions holds. Therefore, there is a significant difference between the historical volatility and implied risk-neutral volatility, mainly because if the model has any additional risk components (e.g. jump risk, or stochastic volatility), or any imperfection, or if the continuous trading is impossible, then the Black-Scholes theory will not be true. More general theories will tell us that in equilibrium the option prices should be computed under a risk-adjusted probability[1]. Therefore, all the implied parameters from options will be risk-adjusted parameters, not the original physical parameters. Hence, the historical volatility will be quite different from implied volatility, as the latter may depend on investor's utility functions and other equilibrium parameters.

When should we use implied parameters and when should we use historical parameters? If we want to compute derivative prices, then we should use appropriate implied parameters. If we want to calculate the present values of expected future cash flows or carry out scenario analysis to calculate potential future loss from defaults, then we should choose historical parameters.

Note that one advantage of using implied default probabilities is that you can get the values from market security prices frequently, unless historical default probabilities, which, for example, are reported by rating agencies a few times per year. However, one main disadvantage of using implied default probabilities is that different securities give different implied default probabilities; one has to figure out which one to use.

As we shall see implied default probabilities are in general much bigger than corresponding historical default probabilities. The following table is a typical example, with the historical default intensities computed from Moody's historical default data as given previously, and implied default intensity computed from bond prices from Merrill Lych.

[1] In general, there will not be a unique risk-neutral probability. The risk-adjusted probability will be a particular risk-neutral probability.

Rating	7-year Average Historical Default Intensity (%)	7-year Implied Default Intensity (%)	Ratio
Aaa	0.04	0.67	16.8
Aa	0.06	0.78	13
A	0.13	1.28	9.8
Baa	0.47	2.38	5.1
Ba	2.47	5.07	2.1
B	7.69	9.02	1.2
Caa	16.90	21.30	1.3

For example, for the A-rate class, it follows from the table of Section 2.1 that we have

$$P(\tau < 7) = 0.91\%.$$

Then

$$P(\tau \geq 7) = 1 - P(\tau < 7) = 1 - 0.0091.$$

So, the computation for the historical default intensity is given by

$$-\frac{1}{7}\ln(1 - 0.0091) = 0.13\%,$$

by estimating the average hazard rate $\bar{\lambda}(t)$ in

$$P(\tau \geq t) = \exp\left\{-\int_0^t \lambda(s)ds\right\} =: \exp\{-\bar{\lambda}(t)t\}.$$

There are several reasons for the big difference. (1) Since default probabilities are related to corporate/foreign bonds, investors demand extra returns for liquidity, as corporate/foreign bonds are relatively illiquid comparing to default-free bonds, e.g. U.S. treasury bonds. (2) Historical default probabilities are estimated within a given time periods. Bond holders may want to allow for extra rooms for the range outside the given time periods. (3) Unlike stocks, bond portfolios are in general not as diversified as stock portfolios, whence investors may demand extra returns for bearing undiversified risk.

6 Estimating Risk-Adjusted Default Probabilities from Bond Prices

To estimate default probabilities from bond prices, one needs to make assumptions about recovery rates.

6.1 Recovery

How much the debt holders can recover once a default occurs is matter of arts rather than science, as the whole recovery process involves complicated legal and financial issues. However, there are some well-established empirical evidences. For example, for high-tech companies (such as internet and biotech companies), the recovery values are generally low, as they have few tangible assets; while for asset-rich (e.g. such as airlines and utility) industries, the recovery values are generally high. There are also some empirical evidences supporting that the higher historical default probabilities the lower recovery rates.

In terms of modelling, there are three popular recovery assumptions. The first is the recovery at a fraction of par value. (For example, if a bond promises to pay \$1 at maturity, then the par value is \$1.) The second is recovery at a fraction of the market value of the bond just before default. The third is recovery at a fraction of the value of otherwise identical default-free bond.

The first one is probably the mostly observed in practice, although the other two are also quite common. The last two approaches may lead to better analytical tractability. We shall use all of the three assumptions in this course.

The recovery values also depend on the seniority of the debt. Assuming recovery at a fractional par value, the historical recovery rates from 1982 to 2003 computed by Moody's are given in the following table. One observation from the table is that the default cost is quite high. Even for senior secured debts, the loss is almost 50%.

Class	Average Recovery Rate (%)
Senior secured	51.6
Senior unsecured	36.1
Senior subordinated	32.5
Subordinated	31.1
Junior subordinated	24.5

Of course the above table only gives the average recovery rates. For two firms, one being asset rich and the other not, the senior secured debts of the two firms will have different recovery rate, although the debts belongs to the same class.

6.2 Implied Default Probabilities from Bond Prices

To illustrate the method, consider a corporate bond with maturity 5 years, and a coupon payment 6% (paid semiannually), with the par value $100. Assume that the current yield of the bond is 7% per year (with continuous compounding), while the current yield on the otherwise identical risk-free bond is 5% (with continuous compounding). Furthermore, assume the recovery is 40% at the par value.

Then we have the following table, assuming the default can only happen at times 0.5, 1.5, 2.5, 3.5, 4.5 years, and the one-year default probabilities are the same.

Time (Years)	Default Intensity (per year)	Recovery Value ($)	Risk-Free Value ($)	Loss Given Default ($)	Discount Factor	PV of Expected Loss ($)
0.5	P	40	106.73	66.73	0.9753	65.08P
1.5	P	40	105.97	65.97	0.9277	61.20P
2.5	P	40	105.17	65.17	0.8825	57.52P
3.5	P	40	104.34	64.34	0.8395	54.01P
4.5	P	40	103.46	63.36	0.7985	50.67P
					Total:	288.48P

For example, the calculation for 3.5 is as follows.

(1) The risk-free value of the future cash flow at year 3.5 is given by

$$3 + 3e^{-0.05*0.5} + 3e^{-0.05*1} + 103e^{-0.05*1.5} = 104.34,$$

as the payments at 3.5, 4, 4.5, 5 years are 3, 3, 3, 103, respectively.

(2) The loss if the default happens between year 3.5 and 4.5 is given by

$$104.34 - 40 = 64.34.$$

(3) The discount factor is

$$e^{-3.5*0.05} = 0.8395.$$

(4) The discounted loss if the default happens between year 3.5 and 4.5 is given by

$$64.34 * 0.8395 = 54.01.$$

Now since the yield is 7% the corporate bond price is

$$3e^{-0.07*0.5} + 3e^{-0.07*1} + 3e^{-0.07*1.5} + 3e^{-0.07*2} + 3e^{-0.07*2.5}$$
$$+3e^{-0.07*3} + 3e^{-0.07*3.5} + 3e^{-0.07*4} + 3e^{-0.07*4.5} + 103e^{-0.07*5} = 95.34.$$

Similarly, the default-free bond price is

$$3e^{-0.05*0.5} + \cdots + 103e^{-0.05*5} = 104.09.$$

Setting the difference between the corporate bond and default-free bond equal to the PV of the expected loss yields

$$104.09 - 95.34 = 288.48P,$$

i.e. the implied default intensity (or implied one-year default probability given no default yet) is

$$P = 3.03\%.$$

The above approach can be easily generalized to the case of different default probabilities for a given year, and with default happens more frequently than a year.

Exercise 2. A company has issued 3- and 5-year bonds with a coupon payment of 4% per year payable annually. The yields on the bonds (in terms of continuous compounding) are 4.5% and 4.75%, respectively. Assume the risk-free rate is 3.5% for all maturities, and the recovery value is 40% of the par value. Also assume default can only take place halfway through each year, i.e. at year 0.5, 1.5, 2.5, etc. Compute P_1 and P_2, which are the implied default intensities per year for years 1 to 3 and for years 4 and 5, respectively.